

TrES-5b

R-1720

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_160914 · created 2026-08-02T16:27:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457645.8741 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.053
Transit depth (Rp/Rs)^2	3.34 +/- 1.96 [%]
Orbital Inclination (inc)	89.06 +/- 2.88
Airmass coefficient 1 (a1)	0.97 +/- 0.3
Airmass coefficient 2 (a2)	0.025 +/- 0.091
Scatter in the residuals of the lightcurve fit is	31.25 %
Best Comparison Star	#1 - [446, 78]
Optimal Aperture	4.12
Optimal Annulus	4.8
Transit Duration (day)	0.0871 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.183 ± 0.053 vs 0.142 (deviation 0.54σ)
Duration fit vs published	0.0871 d vs 0.0758 d (deviation 1.13σ)
Mid-time O–C	3.4 min
Depth SNR	2.4
Residual scatter	31.25 %

Light curve

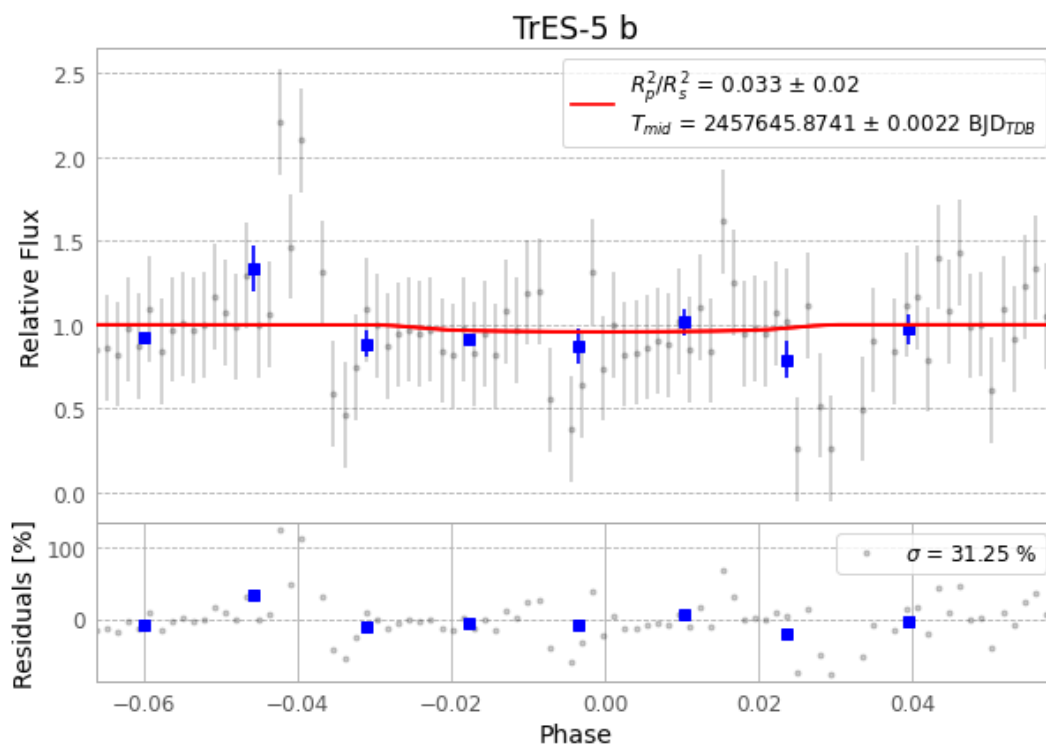


Figure 1 — FinalLightCurve_TrES-5 b_2016-09-13.png (SHA-256 434c35bbeafd4667...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [248, 279], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 539b588b5188d58c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

105 FITS frames (69 MB), TRES-5160914063312.FITS → TRES-5160914114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest.

Artifacts: bench-TRES-5-160914.log (2a369ca4612f...), FinalLightCurve_TrES-5 b_2016-09-13.png (434c35bbeafd...), FinalLightCurve_TrES-5 b_2016-09-13.csv (75d32bfa6583...)

Compute resources

Wall time 260.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 16:27Z.